

Ella Yee

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EDUCATION

Columbia University in the City of New York

New York, NY

B.S. in Biomedical Engineering, Minor in Computer Science

September 2024 - May 2028

- Awards: Egleston Scholar (Top 1% of Engineering Applicants), SURF Fellow, Dean's List
- Coursework: Artificial Intelligence, Computing for Engineers (Python), Data Structures (Java), Advanced Programming (Bash, C), Discrete Math, Multivariable Calculus, Differential Equations, Linear Algebra, Biology
- GPA: 4.2/4.0

SKILLS

- **Languages:** Python, R, Java, Bash/Unix, C
- **Libraries & Tools:** PyTorch, scikit-learn, NumPy, SciPy, Scanpy, AnnData, DESeq2, edgeR, Git
- **Computational Biology:** bulk RNA-seq, scRNA-seq, spatial transcriptomics, cell trajectory analysis
- **Experimental:** PCR, molecular cloning, DNA purification, electrophoresis, immunohistochemistry, histology

EXPERIENCE

ImmuneBridge

San Francisco, CA

Computational Biology Intern

May - August 2026

- Designed computational pipeline for drug-target prioritization integrating transcriptomic analysis, molecular docking, and structure-activity relationship (SAR) data across 50+ proteins and 90+ compounds
- Developed interactive dashboard to compare candidate targets and prioritize experimental validation
- Created automation tools for company-wide workflows (invoice processing, email routing, paper formatting, etc.)

Columbia Engineering

New York, NY

Research Intern, Azizi Lab

September 2025 - Present

- Build Python pipelines for clustering, visualization, and trajectory analysis of scRNA-seq/spatial transcriptomics
- Apply deep generative models (Decipher, Starfish) to identify spatial and cellular dynamics in human GVHD
- Integrate transcriptomic, histological, and clinical data across patient samples to characterize plasma cell states

Columbia University Irving Medical Center

New York, NY

Research Fellow (SURF Program) & Intern, Abate-Shen Lab

September 2024 - August 2025

- Conducted differential expression analysis (R) of RNA-seq datasets from genetically-engineered mouse models
- Evaluated 20+ organ-on-a-chip platforms as ex vivo models, compared engineering constraints and scalability

Stanford Cancer Institute

Stanford, CA

Research Intern, Sunwoo Lab

June 2023 - August 2024

- Performed molecular cloning, DNA purification, and PCR to evaluate constructs for CAR-NK immunotherapy
- Designed CRISPR-Cas9 gene knockouts using lentiviral transduction and quantified gene expression by qPCR

NewMind Discovery & Harker Labs

San Jose, CA

Independent Researcher

August 2020 - May 2024

- Built machine learning pipelines to analyze sequencing and image data across 4 research projects using Python/R
- Investigated mechanisms and candidate biomarkers for diabetic retinopathy, autism spectrum disorder, and cancer

UC Davis Young Scholars Program (YSP)

Davis, CA

Research Scholar, Korf Lab

June - July 2022

- Selected for 6-week residential research program at UC Davis Genome Center (<12% acceptance rate)
- Developed Python/Unix algorithms to identify ultraconserved genomic elements (UCEs) across 4 plant species using JBrowse and NIH BLAST, received "Summer Slam" award for best oral presentation

LEADERSHIP, PUBLICATIONS, & AWARDS

- Society of Women Engineers Treasurer, Columbia Undergraduate Science Journal Board, Claude Builder Club
- *ACM-BCB*, "Identifying Clock Gene Signatures Through Gene Expression Analysis for Cancer Therapy"
- *ICBCB (IEEE)*, "Identifying Neural Patterns and Biomarkers of ASD through Multi-phase Resting-State Functional MRI Analysis," DOI: 10.1109/ICBCB57893.2023.10246705
- 2x Regeneron International Science and Engineering Fair (ISEF) Grand Award, Science Talent Search Top 300
- British Biology Olympiad Gold Medalist; USA Biology Olympiad National Semifinalist
- 4x American Invitational Mathematics Exam (AIME) Qualifier, Math Prize for Girls (MP4G)